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Email: panigrahis@nitrkl.ac.in

Dr. Sibarama Panigrahi

Associate Professor



Academic

Ph. D. in CSE, *VSSUT (UCE) Burla, Odisha (Oldest Govt. Engineering College)* 2019
M. Tech. in CSE, *VSSUT (UCE) Burla, Odisha (Oldest Govt. Engineering College)* (**University Silver Medal**) 2013
B. Tech. in CSE, *Biju Patnaik University of Technology* 2009

Experience

Associate Professor, National Institute of Technology, Rourkela, Odisha (An Institute of National Importance) 14/10/2026 - Till date
Assistant Professor Grade-I (AGP 8000), National Institute of Technology, Rourkela, Odisha (An Institute of National Importance) 03/04/2023 - 14/10/2026
Assistant Professor, SUIIT, Sambalpur University, Burla, Odisha (State Govt. Funded Institute) 26/09/2016 - 31/03/2023

Research and Teaching Interests

Research: Time Series Forecasting, Data Science, Soft Computing, Machine Learning
Teaching: Programming for Problem Solving, Data Structure and Algorithms, Database Engineering, Data Science, Soft Computing, Time Series Analysis, Introduction to Artificial Intelligence and Machine Learning, Deep Learning.

Projects/Grants Undertaken (Total Amount: Rs.88.566 Lakh)

Design and Development of Large Scale Time Series Forecasting Methods using Deep Learning Techniques	4.500 Lakh, (**Role: Sole PI**), Duration: Sep 2021– Aug 2023	Odisha State Higher Education Council
Studies on Deep Learning Models for Crude Oil Price Forecasting	22.077 Lakh, (**Role: Sole PI**), Duration: Mar 2022– Dec 2025	Core Research Grant Scheme, SERB (Presently ANRF)
Post-Disaster Crop Damage Assessment: A Step Towards Sustainable Agriculture	61.9890 Lakh (**Role: Co-PI**), Duration: Nov 2025– Sep 2030	PAIR Scheme, ANRF

Academic Achievements

1. **D**eveloped an electricity load forecasting software employing machine learning techniques for Western Odisha through the project sponsored in OURIIP-2020 Scheme. This software is presently being used by TPWODL for power purchasing and planning of Western Odisha. **F**or this, received letter of appreciation from TPWODL and Vice-Chancellor, Sambalpur University.
2. **Q**ualified GATE-2012 (Score: 593, Marks:52.33).
3. **Q**ualified UGC NET in Computer Science and Applications.
4. **Associate Editor** of Plos One Journal
5. **Associate Editor** of Scientific Reports Journal.
6. Senior Member, IEEE

Journal Publication (Total: 27, Cumulative Impact Factor:100.4, Q1:#15, Q2:#12) and Patents

- **Elsevier:08–** Engineering Applications of Artificial Intelligence (3), Information Sciences (2), Pattern Recognition Letters (1), Computational and Theoretical Chemistry (1), Environmental Modelling & Software (1).
- **Springer:09–** Soft Computing (1), Neural Computing & Applications(1), Wireless Personal Communications (1), Solar Physics (1), Asia-Pacific Financial Markets(1), Computational Economics(1), Arabian Journal for Sc. & Engineering (3).
- **Wiley:05–** Expert Systems (1), Journal of Forecasting(1), International Transactions on Electrical Energy Systems (2), Optimal Control Applications and Methods (1).
- **Taylor and Francis:03–** Applied Artificial Intelligence (2), Journal of Biopharmaceutical Statistics (1).
- **American Chemical Society:01–** Journal of Chemical Information and Modeling(1) **Others:**1.
- **Patents:** 01 (Granted) + 01 (Published)

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List of SCI Indexed Journal Publications

1. Sibarama Panigrahi, and H. S. Behera*. "A hybrid ETS–ANN model for time series forecasting." Engineering Applications of Artificial Intelligence 66 (2017): 49-59. https://doi.org/10.1016/j.engappai.2017.07.007
2. Sibarama Panigrahi*. "A novel hybrid chemical reaction optimization algorithm with adaptive differential evolution mutation strategies for higher order neural network training." International Arab Journal of Information Technology 14(2017): 18-25.
3. Sibarama Panigrahi, and H. S. Behera*. "Time Series Forecasting Using Differential Evolution-Based ANN Modelling Scheme." Arabian Journal for Science and Engineering 45(2020): 11129-11146. https://doi.org/10.1007/s13369-020-05004-5
4. R. M. Pattanayak, Sibarama Panigrahi, and H. S. Behera*. "High-order fuzzy time series forecasting by using membership values along with Data and Support Vector Machine." Arabian Journal for Science and Engineering 45 (2020): 10311-10325. https://doi.org/10.1007/s13369-020-04721-1
5. Sibarama Panigrahi, and H. S. Behera*. "A study on leading machine learning techniques for high order fuzzy time series forecasting." Engineering Applications of Artificial Intelligence 87 (2020): 103245.https://doi.org/10.1016/j.engappai.2019.103245
6. Sibarama Panigrahi*, R. M. Pattanayak, P. K. Sethy, S. K. Behera. "Forecasting of Sunspot Time Series Using a Hybridization of ARIMA, ETS and SVM Methods." Solar Physics 296 (2021): 1-19. https://doi.org/10.1007/s11207-020-01757-2
7. S. K. Purohit, Sibarama Panigrahi*, P. K. Sethy, and S. K. Behera. "Time series forecasting of price of agricultural products using hybrid methods." Applied Artificial Intelligence 35 (2021) 1388-1406. https://doi.org/10.1080/08839514.2021.1981659
8. R. M. Pattanayak, H. S. Behera*, and Sibarama Panigrahi. "A novel probabilistic intuitionistic fuzzy set based model for high order fuzzy time series forecasting." Engineering Applications of Artificial Intelligence 99 (2021): 104136. https://doi.org/10.1016/j.engappai.2020.104136
9. S. K. Behera, P. K. Sethy*, S. K. Sahoo, Sibarama Panigrahi, and S. C. Rajpoot, "On-tree fruit monitoring system using IoT and image analysis." Concurrent Engineering 29(2021) :6-15. https://doi.org/10.1177/1063293X20988395
10. P. K. Sethy, C. Pandey*, M. R. Khan, S. K. Behera, K. Vijaykumar, and Sibarama Panigrahi,"A cost-effective computer-vision based breast cancer diagnosis." Journal of Intelligent & Fuzzy Systems 41(2021):5253-5263. https://doi.org/10.3233/JIFS-189848
11. M. K. Kar, S. Kumar, A. K. Singh, and Sibarama Panigrahi*, "Reactive power management by using a modified differential evolution algorithm." Optimal Control Applications and Methods 44(2021): 967-986. https://doi.org/10.1002/oca.2815
12. M. K. Kar, S. Kumar, A. K. Singh, and Sibarama Panigrahi*, "A modified sine cosine algorithm with ensemble search agent updating schemes for small signal stability analysis." International Transactions on Electrical Energy Systems 31(2021): e13058. https://doi.org/10.1002/2050-7038.13058
13. R. M. Pattanayak, H. S. Behera*, and Sibarama Panigrahi. "A non-probabilistic neutrosophic entropy-based method for high-order fuzzy time-series forecasting." Arabian Journal for Science and Engineering 47 (2022): 1399-1421. https://doi.org/10.1007/s13369-021-05718-0
14. R. Pradhan, Sibarama Panigrahi, and P. K. Sahu*, "Implementation of gradient gravitational search algorithm towards conformational search." Computational and Theoretical Chemistry 1208(2022): 113550.https://doi.org/10.1016/j.comptc.2021.113550
15. R. Yelchuri, J. K. Dash*, P. Singh, A. Mahapatro, and Sibarama Panigrahi, "Exploiting deep and hand-crafted features for texture image retrieval using class membership." Pattern Recognition Letters 160(2022):163-171. https://doi.org/10.1016/j.patrec.2022.06.017
16. M. K. Kar, S. Kumar, A. K. Singh, S. Panigrahi, and M. Cherukuri*, "Design and analysis of FOPID-Based damping controllers using a modified grey wolf optimization algorithm." International Transactions on Electrical Energy Systems 339630,(2022):1-31. https://doi.org/10.1155/2022/5339630
17. S. S. Pradhan, Sibarama Panigrahi*, S. K. Purohit, and J. K. Dash, "Study and development of hybrid and ensemble forecasting models for air quality index forecasting", Expert Systems 40(2023): e13449. https://doi.org/10.1111/exsy.13449

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18. R. M. Pattanayak*, H. S. Behera, and Sibarama Panigrahi. "A Novel High Order Hesitant Fuzzy Time Series Forecasting by using mean Aggregated Membership value with Support Vector Machine." Information Sciences 626 (2023): 494-523. https://doi.org/10.1016/j.ins.2023.01.075

19. R. Pradhan, Sibarama Panigrahi, and P. K. Sahu*, "Conformational search for the building block of proteins based on the gradient gravitational search algorithm (ConfGGS) using force fields: CHARMM, AMBER, And OPLS-AA." Journal of Chemical Information and Modeling 63(2023): 670–690. https://doi.org/10.1021/acs.jcim.2c01398

20. K. Das, S. Das, and Sibarama Panigrahi*,"Energy-efficient forecasting of temperature data in sensor cloud system using a hybrid SVM-ANN method." Wireless Personal Communications 129(2023): 2929–2944. https://doi.org/10.1007/s11277-023-10265-y

21. G. Shial, S. Sahoo, and Sibarama Panigrahi*,"An enhanced GWO algorithm with improved explorative search capability for global optimization and data clustering." Applied Artificial Intelligence 37(2023): e2166232. https://doi.org/10.1080/08839514.2023.2166232

22. T. K. Behera, S. Sathia, Sibarama Panigrahi, and P. K. Naik*, "Revolutionizing cardiovascular disease classification through machine learning and statistical methods." Journal of Biopharmaceutical Statistics (2024):1-23. https://doi.org/10.1080/10543406.2024.2429524

23. S. S. Pradhan and Sibarama Panigrahi*,"A study and development of high-order fuzzy time series forecasting methods for air quality index forecasting." Journal of Forecasting 43(2024): 1-24. https://doi.org/10.1002/for.3153

24. A. P. Padhy, Sibarama Panigrahi*, V. P. Singh, and P. Pratyasha, "Model order reduction for SISO and MIMO system using improved adaptive differential evolution algorithm", Soft Computing, (2024):1-20. https://doi.org/10.1007/s00500-023-09489-8

25. S. K. Purohit and Sibarama Panigrahi*,"Novel deterministic and probabilistic forecasting methods for crude oil price employing optimized deep learning, statistical and hybrid models." Information Sciences 658(2024): 120021. https://doi.org/10.1016/j.ins.2023.120021

26. S. S. Pradhan, S. Panigrahi*, S. K. Purohit. ”A Novel Series-Parallel Hybrid Method employing Fusion of Deep Features for Air Quality Index Forecasting.” Environmental Modelling & Software 204 (2026): 107107. https://doi.org/10.1016/j.envsoft.2026.107107

27. C. R. Padhan, S. Senthil, S. Panigrahi*, B. Sahoo, "An Optimized VMD-Based Deterministic and Probabilistic Crude Oil Price Forecasting Method Employing Machine Learning Models." Computational Economics (2026). https://doi.org/10.1007/s10614-026-11436-2

List of Scopus Indexed Journal Publications

1. K.K. Sahu, Sibarama Panigrahi, and H. S. Behera*. "A novel chemical reaction optimization algorithm for higher order neural network training." Journal of Theoretical & Applied Information Technology 53(2013):402-409.

2. Y. Karali, Sibarama Panigrahi, and H. S. Behera*. "A Novel Differential Evolution based Algorithm for Higher Order Neural Network Training." Journal of Theoretical & Applied Information Technology 56(2013):355-361.

3. Sibarama Panigrahi, and H. S. Behera*. "A computationally efficient method for high order fuzzy time series forecasting." Journal of Theoretical & Applied Information Technology 96 (2018): 7215-7226.

4. G. Shial, S. Sahoo, and Sibarama Panigrahi*. "A Nature Inspired Hybrid Partitional Clustering Method Based on Grey Wolf Optimization and JAYA Algorithm." Computer Science 24 (2023):1-45. https://doi.org/10.7494/csci.2023.24.3.4962

5. G. Shial, C. R. Tripathy, S. Sahoo, and Sibarama Panigrahi*. "A sine-cosine algorithm blended grey wolf optimisation algorithm for partitional clustering." Int. J. Computational Vision and Robotics 15(2025):198-232. https://doi.org/10.1504/IJCVR.2023.10059975

6. S. K. Purohit, and Sibarama Panigrahi*. "Decomposition-based hybrid methods employing statistical, machine learning, and deep learning models for crude oil price forecasting." Neural Computing and Applications 2025:1-46. https://doi.org/10.1007/s00521-025-11178-6 [IF:4.5]

7. G. Yaswanth, H. Lohith, M. Jerusha, and Sibarama Panigrah*. "A Study on Forecasting of Crude Oil Prices Employing Optimized Deep Learning Models and LUBE Method." Asia-Pacific Financial Markets (2025):1-46. https://doi.org/10.1007/s10690-025-09554-2

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List of International Conference Publications

1. Sibarama Panigrahi*, B. Rath, and P. S. Kumar,"A hybrid CRO-K-means algorithm for data clustering." in Computational Intelligence in Data Mining, Vol.3, pp.627-639, Springer 2014. https://doi.org/10.1007/978-81-322-2202-6_57
2. Sibarama Panigrahi*, H.S. Behera, and A. Abraham, "A Fuzzy Filter Based Hybrid ARIMA-ANN Model For Time Series Forecasting." in Proceedings Of The Eighth International Conference On Soft Computing And Pattern Recognition (SoCPaR 2016), pp.592–601, Springer 2017. https://doi.org/10.1007/978-3-319-60618-7_58
3. Sibarama Panigrahi* and H. S. Behera,"An Adaptive Fuzzy Filter-Based Hybrid ARIMA-HONN Model For Time Series Forecasting", in Computational Intelligence In Data Mining: Proceedings Of The International Conference On CIDM 2017, vol.711, pp.841–850, Springer 2018. https://doi.org/10.1007/978-981-10-8055-5_74
4. Sibarama Panigrahi* and H.S. Behera,"Time series forecasting using a hybrid Jaya-FLANN Model." in 2018 International Conference on Recent Innovations in Electrical, Electronics & Communication Engineering (ICRIEECE), pp.3402-3407, IEEE, Bhubaneswar, India 2018. https://doi.org/10.1109/ICRIEECE44171.2018.9008916
5. R. M. Pattanayak, H. S. Behera*, and Sibarama Panigrahi,"A Multi-step-ahead fuzzy time series forecasting by using hybrid chemical reaction optimization with Pi-sigma higher-order neural network." in Computational Intelligence in Pattern Recognition, pp.1029–1041, Springer 2019. https://doi.org/10.1007/978-981-13-9042-5_88
6. Sibarama Panigrahi* and H. S. Behera,"Fuzzy time series forecasting: a survey", in Computational Intelligence In Data Mining: Proceedings Of The International Conference On ICCIDM 2018, vol.990, pp.641-651, Springer 2019. https://doi.org/10.1007/978-981-13-8676-3_54
7. R. M. Pattanayak*, H.S. Behera, and Sibarama Panigrahi,"A Novel Hybrid Differential Evolution-PSNN For Fuzzy Time Series Forecasting." in Computational Intelligence in Data Mining, vol.990, pp.675–687, Springer 2019. https://doi.org/10.1007/978-981-13-8676-3_57
8. G. Shial, S. Sahoo, and Sibarama Panigrahi*,"Community detection and disease identification using meta-heuristic based clustering methods", in 2022 IEEE India Council International Subsections Conference (INDISCON), pp.1-6, IEEE, Bhubaneswar, India 2022. https://doi.org/10.1109/INDISCON54605.2022.9862931
9. G. Shial, S. Sahoo, and Sibarama Panigrahi*,"Identification and analysis of breast cancer disease using swarm and evolutionary algorithm", in 2022 IEEE Region 10 Symposium (TENSYMP), pp.1-6, IEEE, Mumbai, India 2022. https://doi.org/10.1109/TENSYMP54529.2022.9864514
10. S. S. Pradhan and Sibarama Panigrahi*,"Studies on machine learning techniques for multivariate forecasting of Delhi air quality index", in Advances in Data-Driven Computing and Intelligent Systems. ADCIS 2022., vol.698, pp.133–146, Springer, August 2023. https://doi.org/10.1007/978-981-99-3250-4_10
11. S. S. Pradhan and Sibarama Panigrahi*,"Delhi air quality index forecasting using statistical and machine learning models", in AIP Conference Proceedings, vol.2705, no.1, pp.020002, American Institute of Physics, June 2023. https://doi.org/10.1063/5.0133357
12. G. Shial, C. Tripathy, Sibarama Panigrahi, and S. Sahoo,"An Improved GWO algorithm for data clustering", in Computing, Communication and Learning. CoCoLe 2022, vol.1729, pp.79–90, Springer 2023. https://doi.org/10.1007/978-3-031-21750-0_7
13. S. K. Purohit and Sibarama Panigrahi*,"Ranking optimised statistical models for time series forecasting of crude oil price", in Computing, Communication and Intelligence, vol.1, Taylor and Francis, November 2024. https://doi.org/10.1201/9781003581215-36
14. S. K. Purohit and Sibarama Panigrahi*,"Forecasting Crude Oil Prices: A Machine Learning Perspective", in International Conference on Computing, Communication and Learning, vol.1892, pp.15-26, Springer, March 2024. https://doi.org/10.1007/978-3-031-56998-2_2
15. Kadiyam, Aswin Tirumala Someswar, Chinmaya Ranjan Padhan, and Sibarama Panigrahi. "A Study on Object Detection using Faster R-CNN, Retina net, RPN_R_50_FPN, RPN_R_50_C4." In 2024 International Conference on Intelligent Computing and Sustainable Innovations in Technology (IC-SIT), pp. 1-6. IEEE, 2024. https://doi.org/10.1109/IC-SIT63503.2024.10862488
16. C. Jayanth, and Sibarama Panigrahi. "Classification of Pigmented Skin Lesions using Early Fusion of Multi-Deep Features and Support Vector Machine." In 2024 International Conference on Intelligent Computing and Sustainable Innovations in Technology (IC-SIT), pp. 1-6. IEEE, 2024. https://doi.org/10.1109/IC-SIT63503.2024.10862008

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PhD Guidance (Completed:02, Ongoing:08)

Name of the Scholar	Thesis Title	Institute / University	Status
Sushree Subhaprada Pradhan	Forecasting of Air Quality Index using Machine Learning Techniques	Sambalpur University	Awarded [21 November 2024]
Sourav Kumar Purohit	Studies on Machine Learning Techniques for Crude Oil Price Forecasting	Sambalpur University	Awarded [25 October 2025]

M.Tech Guidance (Completed: 17, Ongoing: 04)

Name of the Scholar	Thesis Title	Institute / University	Status
P. Santosh Kumar	Experimental Study of Complex Valued Higher Order Neural Network for Pattern Classification	Berhampur University	Awarded [October 2014]
Balaram Rath	A Novel Evolutionary Clustering Algorithm and its Performance Analysis	Berhampur University	Awarded [October 2014]
Priyanka Behera	Fuzzy Time Series Forecasting	SUIIT, Sambalpur University	Awarded [June 2018]
Kunal Mishra	Crude Oil Price Forecasting using Artificial Neural Network	SUIIT, Sambalpur University	Awarded [June 2019]
Shreemat Kumar Dash	A Machine Learning Approach: Stock Price Prediction using ANN	SUIIT, Sambalpur University	Awarded [June 2019]
Aparna P. Mishra	Approaches for Data Security through Modified AES Algorithm	SUIIT, Sambalpur University	Awarded [June 2020]
Chandrika Kumari	Jelly Fish Swarm Algorithm - Feature Selection using Wrapper Method	SUIIT, Sambalpur University	Awarded [June 2020]
Manas Kumar Patel	Studies on Machine Learning Models for Predicting Hard Disc Drive Failure	SUIIT, Sambalpur University	Awarded [June 2021]
Mohit Rathore	Enhancing Software Quality by Predicting Defects using Machine Learning Techniques	NIT Rourkela	Awarded [June 2024]
Chowdam Jayanth	Air Quality Index Forecasting using Optimized Ensemble and Hybrid Methods	NIT Rourkela	Awarded [June 2025]
Sunrit Banerjee	Studies On Probabilistic Forecasting Methods Employing Deep Learning Models	NIT Rourkela	Awarded [June 2025]
Tapas Manna	Hybrid Fuzzy Time Series Forecasting Integrating Machine Learning Models	NIT Rourkela	Awarded [June 2025]
Aman Maurya	Multivariate Time Series Classification through Image Transformation using Deep Learning	NIT Rourkela	Awarded [June 2025]
Raman Kumar Pal	Crude Oil Price Prediction using Machine Learning Techniques	NIT Rourkela	Awarded [June 2025]
Prakash Kumar Garg	Adaptive and Efficient Deep Learning Frameworks for Computer Vision Applications	NIT Rourkela	Awarded [June 2026]
Krishna	Multivariate Time Series Classification through Image Transformation using Deep Learning	NIT Rourkela	Awarded [June 2026]
Sagar Das	Probabilistic Hierarchical Graph Neural Network for Aggregation-Coherent Time Series Forecasting	NIT Rourkela	Awarded [June 2026]

Reviewer

1. Reviewer of Anusandhan National Research Foundation (ANRF) Projects.
2. Reviewer of Board of Research in Nuclear Sciences (BRNS) Projects.
3. **SCI Journals of Elsevier:** Engineering Applications of Artificial Intelligence, Expert Systems with Applications, Information

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Sciences, Information Fusion, Pattern Recognition, Neurocomputing, Technological Forecasting & Social Change, EconomiA, Energy, Heliyon, MethodsX.

4. **SCI Journals of IEEE:** IEEE Transactions on Knowledge and Data Engineering, IEEE Transactions on Emerging Topics in Computing, IEEE Access.
5. **SCI Journals of Springer:** Soft Computing, Evolutionary Intelligence, Iranian Journal of Science and Technology, Transactions of Civil Engineering, International Journal of Data Science and Analytics, International Journal of Computational Intelligence Systems, Scientific Reports.
6. **SCI Journals of Wiley:** Journal of Forecasting, International Journal of Communication Systems, Expert Systems.
7. **SCI Journals of Taylor & Francis:** Communications in Statistics: Case Studies, Data Analysis, Computer methods in biomechanics and biomedical engineering, Hydrological Sciences Journal, Journal of Crop Improvement, Network: Computation in Neural Systems.
8. **SCI Journals of Plos:** Plos One.
9. **SCI Journals of SAGE:** Concurrent engineering, research and applications.

List of Short-term Courses / Workshops / Conferences Organized

1. National Workshop on "Neural Network and Deep Learning", 21 Feb -25 Feb 2023 at Sambalpur University Institute of Information Technology. (Convener)
2. Short Term Course on “Recent Trends in Time Series Forecasting using Deep Learning Models”, 25 Dec 2023 - 29 Dec 2023 at NIT Rourkela. (Coordinator)
3. Short Term Course on “Introduction to Data Science using Python”, 20 May 2024 - 24 May 2024 at NIT Rourkela. (Coordinator)
4. National Workshop on "Crisp and Fuzzy Time Series Forecasting using Deep Learning Techniques", 25 Sep 2024 - 29 Sep 2024 at NIT Rourkela. (Convener)

Resource Person

1. Delivered an Invited Talk on “The Art of Writing Research Proposals for Funding” in the Online Webinar, organized by Amity Institute of Information Technology, Amity University Jharkhand, Ranchi, on 09 June, 2023.
2. Delivered an Invited Talk on “The Art of Writing Research Proposals for Funding” in the Online Webinar, organized by Driems University, Cuttack, Odisha, on 24 June, 2023.
3. Delivered an Invited Talk on “Advances in Neural Network and Deep Learning” in the One Week International Workshop on Recent Trends in Mathematics and Computing (IWRTMC2023), organized by VSSUT, Burla, Odisha, on 19 August 2023.
4. Delivered an Invited Talk on “Time Series Forecasting employing Optimized Deep Learning Models” in the Online Workshop on “Artificial Intelligence in Materials Engineering: Theory and Hands-on Practice” organized by Department of Metallurgical and Materials Engineering, National Institute of Technology Rourkela, on 21 September 2023.
5. Delivered an Invited Talk on “Time Series Forecasting using Optimized Deep Learning and Hybrid Models”, in the Seminar organized by IEEE Student Branch, VSST, Burla on IEEE day 30 September 2023.
6. Delivered an Invited Talk on “Advances in Neural Network and Deep Learning”, in the AICTE Sponsored One Week Faculty Development Program on "Exploring the Technological trends in Machine Learning and Artificial Intelligence-Need of the Hour", organized by MVJ College of Engineering, Bangalore on 9 October 2023.
7. Delivered an Invited Talk on “Time Series Forecasting using Optimized Deep Learning, Statistical and Hybrid Models”, in the one week workshop on "Intelligent Computing for Signal Processing Applications", organized by Sambalpur University Institute of Information Technology, Burla, Odisha on 17 January 2024.
8. Delivered an Invited Talk on “Take Control of Your Data” organized by the Institution of Engineers (India), Rourkela Local Centre, on the occasion of Data Protection Day on 29 January 2024.
9. Delivered an Invited Talk on “Time Series Forecasting using Deep Learning Techniques” in the Faculty Development Programme on “INDUSTRY 4.0 AND SMART SYSTEMS” (Funded by SERB. Govt. of India), organized by JIS College of Engineering, Kolkata on 9 February 2024.

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10. Delivered an Invited Talk on “Modern Techniques of Information and Communication Technology (ICT) in Education” in the Second Faculty Development Programme on National Education Policy-2020 Orientation and Sensitization Programme conducted by UGCMalaviya Mission Teacher Training Programme, Sambalpur University 16 February 2024.
11. Delivered an Invited Talk on “Whale Optimization Algorithm and its Application to Deep Learning Based Time Series Forecasting” in the Faculty Development Programme on “Optimization Technique for Engineering Applications”, organized by E&ICT, NIT Warangal on 22 March 2024.
12. Delivered a Pre-Conference Tutorial Talk on “Recent Adavances in Time Series Forecasting using Machine Learning Techniques” in the International Conference on on Machine Learning, IoT and Big Data (ICMIB-2024), Organized by GIET University, Gunupur on 29 March 2024.
13. Delivered an Invited Talk on “Whale Optimization Algorithm and its Application to Deep Learning Based Time Series Forecasting” in the Faculty Development Programme on “Emerging Applications of Optimization Techniques”, organized by E&ICT, KITS Warangal on 12 April 2024.
14. Delivered an Invited Talk on “Modern Techniques of Information and Communication Technology (ICT) in Education” in the UGC-Malaviya Mission Teacher Training Programme organized by North-Eastern Hill University, Shillong on 22 April 2024.
15. Delivered an Invited Talk on “Expert Systems Applications with AI Techniques” in the workshop on “Artificial Intelligence & Machine Learning:Theory & Hands-on Practice (AIML- 2024)” Sponsored by SERB and organized by Department of CSE, NIT Rourkela on 24 June 2024.
16. Delivered an Invited Talk on “Modern Techniques of Information and Communication Technology (ICT) in Education” in the Third Faculty Development Programme on National Education Policy-2020 Orientation and Sensitization Programme conducted by UGCMalaviya Mission Teacher Training Programme, Sambalpur University 14 June 2024.
17. Delivered an Invited Talk and handled a hands-on session on “Dimensionality Reduction Techniques” in the workshop on “Artificial Intelligence & Machine Learning:Theory & Hands-on Practice (AIML- 2024)” Sponsored by SERB and organized by Department of CSE, NIT Rourkela on 27 June 2024.
18. Delivered an Invited Talk on “Time series analysis on weather data” in the Five-day workshop on “IoT and its Applications”, organized by Ewarn System Private Limited on 15 July 2024.
19. Delivered an Invited Talk on “Modern Techniques of Information and Communication Technology (ICT) in Education” in the Fourth Faculty Development Programme on National Education Policy-2020 Orientation and Sensitization Programme conducted by UGCMalaviya Mission Teacher Training Programme, Sambalpur University 25 July 2024.
20. Delivered an Invited Talk on “Deep Learning employing High Performance Computing” in the AICTE sponsored workshop on High Performance Computing & Its Application in AI (HPCAA-24) organized by Silicon University, Bhubaneswar on 9 August 2024.
21. Delivered an Invited Talk on “Time Series Forecasting employing Deep Learning and High Performance Computing” in the AICTE sponsored workshop on High Performance Computing & Its Application in AI (HPCAA-24) organized by Silicon University, Bhubaneswar on 9 August 2024.
22. Delivered an Invited Talk on “Scalability of Deep Learning Models with HPC” in the AICTE sponsored ATAL -Faculty Development Program (ATAL-FDP) on Future Generation Computing with HPC and AI organized by Silicon University, Bhubaneswar on 14 December 2024.
23. Delivered an Invited Talk on “Whale Optimization Algorithm and its Application to Deep Learning Based Time Series Forecasting” in the E & ICT Sponsored Online FDP on AI-Based Problem-Solving Techniques organized by SPEC R&D and Department of CSE, St. Peter’s Engineering College (SPEC), Hyderabad on 17 December 2024.
24. Delivered an Invited Talk on “Problem Solving using Convolutional Neural Network and Transfer Learning” in the E & ICT Sponsored Online FDP on AI-Based Problem-Solving Techniques organized by SPEC R&D and Department of CSE, St. Peter’s Engineering College (SPEC), Hyderabad on 18 December 2024.
25. Delivered an Invited Talk on “Crisp and Fuzzy Time Series Forecasting using Machine Learning Techniques” in the Ministry of Electronics and Information Technology (MeitY) Sponsored Online FDP on From Research to Reality: Translating AI and ML Innovations into Practice organized by School of CSE & IS, VIT Vellore on 27 February 2025.

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Dr. Sibarama Panigrahi

Associate Professor



26. Delivered an Invited Talk on “Crude Oil Price forecasting using Optimized Deep Learning Models” in the Ministry of Electronics and Information Technology (MeitY) Sponsored Online FDP on From Research to Reality: Translating AI and ML Innovations into Practice organized by School of CSE & IS, VIT Vellore on 28 February 2025.
27. Delivered an Invited Talk on “Introduction to Neural Networks and Deep Learning: A Mathematical Perspective” in E & ICT Sponsored Hybrid FDP on Deep Learning in Medical Image Analysis organized by Electronics and ICT Academy, NIT Warangal and NIISIBA Bhubaneswar on 21 April 2025.
28. Delivered an Invited Talk on “Transfer Learning and Popular CNN Models for Image Analysis” in E & ICT Sponsored Hybrid FDP on Deep Learning in Medical Image Analysis organized by Electronics and ICT Academy, NIT Warangal and NIISIBA Bhubaneswar on 22 April 2025.
29. Delivered an Invited Talk on “Feature Engineering and Fusion for Image Classification” in Ministry of Electronics and Information Technology (MeitY) Sponsored Online FDP on “Feature Engineering: The Backbone of Effective AI and ML Solution Applications”, organized by Department of EE, International Institute of Information Technology (IIIT) Bhubaneswar on 13 May 2025.
30. Delivered an Invited Talk on “Reimagining Business Transformation through Time Series Forecasting using AI & ML Techniques ” in the Online Refresher Course in Commerce and Management Conducted By UGC Malavya Mission Teacher Training Centre (Formerly UGC-HRDC), organized by Sambalpur University on 17 August 2025.
31. Delivered 3 Invited Talks on “Introduction to Artificial Intelligence, Machine Learning, and Deep Learning” in the Faculty Development Programme on "Artificial Intelligence and Machine Learning" Conducted By B.P.U.T. Rourkela during 05-09 February 2026.
32. Delivered 2 Invited Talks on “Crisp and Fuzzy Time Series Forecasting” in the Faculty Development Programme on "RECENT TRENDS ON SMART SENSORS AND COMPUTER APPLICATIONS IN ENERGY SECTOR" Conducted By GATE Berhampur on 11 & 17th May 2026.
33. Delivered an Invited Talk on “Whale Optimization Algorithm” in the E&ICT Sponsored Faculty Development Programme on "Emerging Trends in Soft Computing and its Applications" Conducted By NIT Warangal on 10th August 2026.
34. Delivered an Invited Talk on “Recent Developments in Deep Learning Techniques” in the E&ICT, NIT Warangal Sponsored Online Faculty Development Programme on "AI-Driven Transformation in Research: Empowering Learners with Machine Intelligence-Based ICT Tools" Conducted By NIT Warangal in association with GITAM (Deemed to be University), Hyderabad on 10th September 2026.
35. Delivered an Invited Talk on “LLMs for Time Series Forecasting” in the Ministry of Electronics and Information Technology (MeitY) Sponsored Online Faculty Development Programme on "Large Language Models for Generative AI: Foundations, Design and Applications (LLMGAI)" Conducted By NIT Warangal on 16th September 2026.
36. Delivered an Invited Talk on “Introduction to Classification and Clustering Techniques” in the Faculty Development Programme on "Advancement in Soft Computing Techniques for Sustainable Water Management" Conducted By NIT Warangal on 28th October 2026.